

Hyponatraemia: a framework

(based on JAMA 2022;328:280, BMJ 2019;365:l1774, Nephrol Dial Transplant 2014 doi:10.1093/ndt/gfu040, NEJM 2023;389:1499, NICE CKS hyponatraemia and Derbyshire shared care pathology hyponatraemia guideline, both accessed July 2019)

Low sodium (<135) and ANY symptoms: ADMIT – symptoms represent cerebral oedema

(headache, nausea/vomiting, confusion, somnolence, seizure, coma, cardiorespiratory distress)

Sodium <125mmol/L
PROFOUND hyponatraemia

ADMIT

Sodium 125–129mmol/L
MODERATE hyponatraemia

Discuss with medics TODAY

May suggest admission/referral or investigation in primary care. If plan is to investigate in primary care, follow pathway below

Same-day treatment more likely to be needed if:

- Recent surgery
- Self-induced water intoxication
- Known intracranial pathology

Sodium 130–134mmol/L
MILD hyponatraemia

Investigate in primary care

Why investigate mild low sodium?

- Even mild hyponatraemia is associated with cognitive deficits, falls, hospitalisation, readmission and death.
- Chronic hyponatraemia is a risk factor for osteoporosis/fracture and death.
- Mild, well-tolerated, long-standing low sodium may be acceptable: don't treat just to make the number right, BUT do look for reversible or serious causes using pathway below and causes in main article

Investigation in primary care

➤ **Repeat serum sodium** When? No clear guidance but we suggest promptly to see trend (is it rising/falling/stable?)

➤ **Check serum osmolality, urine osmolality and urine sodium (on same urine sample)**

Serum SODIUM: STABLE

➤ **Now look at SERUM OSMOLALITY**

Serum sodium dropping: discuss with medics with a view to ADMISSION

Serum osmolality <275mosm/kg

➤ **Now look at URINE OSMOLALITY**

Serum osmolality ≥275mosm/kg : DISCUSS WITH MEDICS TODAY

(hyperglycaemia is one cause and may be a sign of diabetic emergency)

Urine osmolality >100mosm/kg

➤ **Now look at URINE SODIUM**

Urine osmolality ≤100mosm/kg: consider differential diagnoses: primary polydipsia, high water/low solute intake (e.g. anorexia, elderly with very restricted diet, beer potomania). Refer/treat as appropriate to underlying cause.

Urine sodium ≤30mmol/L

Assess volaemic status

Hypervolaemic

Possible causes: heart failure, liver cirrhosis, renal disease, nephrotic syndrome (may need admission to manage these diseases)

Hypovolaemic

Possible causes: diarrhoea, vomiting, occult diuretics, fluid in third space

ADMIT for IV fluids + solutes

Increasing oral fluids at home risks of worsening hyponatraemia

Urine sodium >30mmol/L

Assess volaemic status

If diuretics/renal disease, consider these as likely cause but consider other causes too:

Hypovolaemic

Possible causes: vomiting, Addison's disease, renal salt wasting, occult diuretics

Euvolaemic

Possible causes: SIAD, secondary adrenal insufficiency, hypothyroid (rare), occult diuretics

Investigate for underlying cause in primary care:

Bloods: U&Es, TFTs, LFTs, glucose/HbA1c, lipids, 9am cortisol, BNP, myeloma screen

Urinalysis (for protein or blood in renal disease), **CXR**, consider CT abdomen/chest/pelvis

Mild, well-tolerated hyponatraemia may be clinically acceptable. If treatment is in patient's best interest:

- Stop non-essential fluids and fluid restrict (to between 0.5 and 1.5L/d)
- Stop medications which may be contributing (see Hyponatraemia article)
- Repeat sodium: e.g. if mild, well and medication altered, suggest repeat at 2w; be guided by clinical presentation
- If not improving, REFER.

If serum sodium worsens to:

<125mmol/L OR symptoms develop: **ADMIT**
125–129mmol/L: discuss with medics

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